

Software Architecture - Software Architecture Design Methods

Time: 15 minutes.

* Indicates required question

1. Email *

2. Full name *

3. Student ID *

4. Architecture sources include ____.*

1 point

Check all that apply.

- ☐ Theft.
- ☐ Method.
- ☐ Intuition.
- ☐ Quality attributes.

5. Select the activities that can be used for designing architecture suggested by the attribute-driven design (ADD) method? * 1 point

Check all that apply.

- ☐ Choose the module to decompose.
- ☐ Choose the architectural drivers from the set of concrete quality scenarios and functional requirements.
- ☐ Choose an architectural pattern that satisfies the drivers.
- ☐ Instantiate modules and allocate functionality from use cases, and represent the results using multiple views.
- ☐ Verify and refine the use cases and quality scenarios and make them constraints for the child modules.
- ☐ Create a prototype.
- ☐ Repeat the steps above for every module that needs further decomposition.

6. The attribute-driven design (ADD) method can be used for creating _____. * 1 point

Check all that apply.

- ☐ architecture of a greenfield system for a mature domain.
- ☐ architecture of a greenfield system for a domain that is novel.
- ☐ architecture for making changes to an existing system (brownfield).
- ☐ architecture of system with constantly changing requirements.
- ☐ architecture of highly experimental system.

7. Which of the following is true of a context diagram? * 1 point

Check all that apply.

- ☐ Must be based on a use case diagram.
- ☐ Must show the network.
- ☐ Describes what is in and what is outside the scope of the software project.
- ☐ Describes how the software being created interacts with elements in its environment.

8. The inputs for creating overall system structure can be ____.*

1 point

Check all that apply.

- ☐ Architectural styles.
- ☐ Reference architectures.
- ☐ Programming languages.
- ☐ Technologies.

9. What are the activities used for designing architecture using viewpoints and perspectives? * 1 point

Check all that apply.

- ☐ Agreeing on scope and context, constraints, and baseline architectural principles.
- ☐ Identifying and engaging stakeholders.
- ☐ Identifying and using architectural scenarios.
- ☐ Using architectural styles and patterns.
- ☐ Producing architectural models.
- ☐ Documenting the architecture.
- ☐ Validating the architecture.

10. Which views are recommended to be created using the viewpoints and perspectives method? * 1 point

Check all that apply.

- ☐ Context view.
- ☐ Functional view.
- ☐ Information view.
- ☐ Concurrency view.
- ☐ Development view.
- ☐ Process view.
- ☐ Deployment view.
- ☐ Operational view.

11. How should an architect decide which views to create? *

1 point

Mark only one oval.

- ☐ Build one for each function.
- ☐ Build one for each of the major concerns.
- ☐ Build one for each of the viewpoints.
- ☐ Build one for each use case.
- ☐ Build one for each actor.

12. Which core perspectives are recommended to be implemented using the viewpoints and perspectives method?

* 1 point

Check all that apply.

- ☐ Security.
- ☐ Performance and Scalability.
- ☐ Availability and Resilience.
- ☐ Evolution.

13. The viewpoints and perspectives method uses perspective as guidelines for achieving a ____.

* 1 point

Mark only one oval.

- ☐ quality attribute
- ☐ view
- ☐ business goal

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